

PAPER_1201 – UQFF 26D Polynomial Origami Downward Projection Axiom

Critical Rule (26D_DOWNWARD_PROJECTION.md:9)

ALL projections run DOWNWARD from 26D.
2D is a reference plane of observation -- NOT a foundation.
WRONG: 2D -> 3D -> 9D -> 26D
CORRECT: 26D -> 9D -> 3D -> 2D

Dimensional Hierarchy (26D_DOWNWARD_PROJECTION.md:22)

26D -- ORIGIN

UA pure energy, SCm injected -> DPM grinding pair (CW-SCm, CCW-UA')
 $F_{U\{26D\}} = U_g + U_m + U_b + SCm/UA + BBDT * Prob_order$
 $E\{26D\} = UA + SCm * DPM_ref + BBDT$

9D -- VOID SYNTHESIS

$\det(M_{\{26 \rightarrow 9\}})$ compactification matrix (9*9 diagonal [d9,0,0 / ...])
Void_synth = $\det(M_{\{26 \rightarrow 9\}}) * (U_g / U_m / U_b) / d3 + F_inert * E\{26D\} + QFP_unique$
QFP_unique = $\pi_{\{n\}} * IG * Wolfram_rules$

3D -- OBSERVABLE MASS

$M = E\{26D\} / c\{26\} * (1 - v_current / v_init) * Prob_order$
 $c\{26\}$ = speed-of-light equivalent in 26D manifold

2D -- REFERENCE PLANE ONLY

Observable slices (CMB, spectra) -- NOT building block

Core 26D Equations (26D_DOWNWARD_PROJECTION.md:54)

$F_{U\{26D\}} = U_g + U_m + U_b + SCm/UA + BBDT * Prob_order$
 $M = E\{26D\} / c\{26\} * (1 - v_current/v_init) * Prob_order$
Void_synth = $\det(M_{\{26 \rightarrow 9\}}) * (U_g/U_m/U_b)/d3 + F_inert * E\{26D\} + QFP_unique$
Triple-root: $x^3 - 3x^2 + 2x = 0 \rightarrow roots = [0, 1, 2]$

26D Invariants in UQFF Axiom Set (__uqff_primitives.py:632)

$_s26_3 = 1.4531e26$
 $_phi_res = PHI_RES$
 $_n_layers = 26$
 $_kappa = 0.0005$
 $_e0 = 1e-20$

Usage in All 10 derive_* (__uqff_primitives.py:641)

derive_alpha_uqff: $base = 1 / (PHI_RES * N_LAYERS * 2\pi) * (1 + 0.001 * Ubi_corr)$
 $Ubi_corr = F_TRZ * RATIO / (S26_3 ** (1/26))$
derive_G_newton: $lambda_cross = (S26_3 ** (1/26)) * (E0/rho) ** (1/3)$
 $proj_factor = (1 + beta_i) / (N_LAYERS * (1 + F_TRZ))$
derive_beta_i: $cycle_avg = 0.5 + 0.5 * \cos(\pi * F_TRZ)$
 $kappa_geom = (RATIO - 1) * (KAPPA * N_LAYERS)$
derive_particle_masses: $omega_base = THZ * (rho_scm/rho_ua) * (S26_3 ** (1/26)) * beta_i$
derive_hbar: $hbar_base = (E0 * S26_3 * PHI_RES) / (c * 26 * 2\pi) * ubi_avg$
derive_condensed...: $amp = S26_3 * PHI_RES * (1 + KAPPA * 1e4)$
derive_habitable...: uses derive_G + derive_beta + derive_rho (all 26D)

DPMVars26D (source172 + VerificationOrchestrator + 26D_DOWNWARD_PROJECTION.n

19*26 arrays: f_UA_prime[19][26], f_SCm[19][26]

S26_3, det(M_26→9), fold signatures, downward projection invariants

evaluate_26D_polynomial(26D inputs) -> 9D/3D/2D outputs

Used in: GeometricCheckpoint, QCalcGeom CP layers, all derive_*, 9-test Derivations Test

26D Polynomial in Quantum Chain Amplification (dpm_vacuum_manifold.py:139)

VDS = Sum((SSQ**n) / n**26 , (n,1,oo)) # = Li_26([SSq])

E_n = [E0 * 10**n for n in 1..26]

rho_vac_energy = sum(f_SCm * E for E in E_n) / V

Cross-References

- 26D_DOWNWARD_PROJECTION.md:11 (ALL projections downward from 26D)
- __uqff_primitives.py:681 (lambda_cross S26_3^{1/26} in G)
- dpm_vacuum_manifold.py:139 (VDS Li_26)
- QCalcGeom.py:33 (poly26_derivative port)
- LEDGER_FIRST_PRINCIPLES_DERIVATIONS.md:150 (DPM layers = 26D polynomial origami)
- WhitepaperGapAnalysis.md:100 (G3 gap closure)